Covered — Final Product Scope, Feature Set, Voice Interface, RAG Corpus and Technology Decision

Product report · version 2.1 · 24 September 2026 Prepared for: the Covered build team · Supersedes version 1 of the same date *Version 2.1 corrects the team composition — the team includes an engineer — and revises the technology recommendation accordingly (section 8).*

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1. What changed in version 2

Change	Reason
Vehicles enter the product; the **two-wheeler joins the MVP**, three- and four-wheelers follow in phase 2	This is where a consumer saves the most on labour for the least risk. A garage charges for a bulb, a fuse or a diagnosis the owner could have done in four minutes
Agricultural machinery scoped as **phase 3**, with the safety spine designed now	Farm machinery is high-injury equipment. It deserves the product, but not before the safety architecture has been proven on easier categories
Industrial and commercial equipment moved to a **separate B2B track**, not the consumer app	Workplace equipment sits under occupational-safety law and employer duty. A consumer DIY app must not instruct a worker to service plant
Voice-first, hands-free operation promoted to a MUST	Correct instinct, and it is what makes vehicles and farm work at all: the user's hands are dirty, greasy or holding a torch
MUST list cut from 12 to 6	The critique is right. Twelve MUSTs is not an MVP, it is a wish list that fails in review. See section 5
Washing machine moved from MVP to phase 2	Content budget. Five categories is the cap, and the two-wheeler earns a place more than a sixth appliance
New section: the **exact RAG corpus**	Section 7
2.1: the team has an engineer; the build recommendation changes from a managed sandbox to an owned repository	Section 8

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2. Scope by phase

Phase	Categories	Trigger to start
1 — MVP / demo day	Mobile phone · TV + DTH set-top box · refrigerator · cooling (air cooler and AC) · **two-wheeler (motorcycle and scooter)**	Now
2 — first expansion	Washing machine · three-wheeler and car · RO purifier · water pump (diagnose and brief only)	MVP passing its evals with real users

Phase	Categories	Trigger to start
3 — agriculture	Tractor and its routine service points · knapsack and power sprayers · drip and micro-irrigation sets · chaff cutter and thresher **safety guidance only** · solar pump sets	Phase 2 stable, agri-safety corpus reviewed by a domain expert
4 — separate track, not this app	Industrial and commercial equipment: generators, compressors, welding sets, forklifts, workshop machines	A B2B product with an employer as the customer and a different liability posture
Always on, never a repair guide	LPG stove · home inverter and battery	Ships with the MVP as the safety and entitlement module

Why the two-wheeler is the right vehicle to start with

It is the vehicle this audience owns; motor-vehicle ownership is 59% of rural households and 68% of urban. Its most common problems are checklist problems, not mechanical ones, and each of them is a billed garage visit today: it won't start because of the kill switch, the side stand or the brake-lever cut-off; a blown fuse; a failed indicator bulb; a slack or dry chain; a loose battery terminal; a tyre at the wrong pressure. None of these needs a tool the owner doesn't have, and none of them is hazardous when done on a cold, parked machine on level ground.

Vehicle scope — allowed and refused

Allowed (phase 1, two-wheeler; phase 2, car and three-wheeler)	Never, at any phase
Dashboard and telltale-light decoding from a photo	Brakes and any hydraulic line
Won't-start checklist: kill switch, side stand, clutch and brake cut-off, fuel, immobiliser, key-fob battery	Airbags and SRS components
Fuse identification and replacement from the fuse-box diagram	EV high-voltage systems and orange cabling
Bulb type lookup and replacement, where accessible without removing panels	Fuel system, injectors, tank
Tyre pressure against the placard, tread and sidewall inspection, puncture assessment	Working under a jacked vehicle, or any jacking on a slope or soft ground
Chain slack and lubrication, air-filter inspection, wiper blades, coolant and oil level reading	Opening a hot radiator, hot engine work
Battery terminal corrosion check on a switched-off vehicle	Jump-starting is **information only**: the app explains the risk and recommends roadside assistance
Service-schedule and warranty check, free-service entitlement, disputed labour charge	Steering, suspension, clutch overhaul

The money story for vehicles is stronger than for appliances: the app tells the owner which items on the invoice are inside the free-service schedule or warranty, and which "recommended" jobs are optional upselling.

Agriculture scope, drafted now for phase 3

Allowed: air-filter cleaning and dust unloading, radiator screen cleaning, draining water from the fuel filter and sedimenter, greasing the listed points, tyre pressures and ballast, battery terminals, sprayer nozzle cleaning and calibration, drip-filter flushing and emitter cleaning, the operator's daily pre-start checklist.

Never: PTO shafts, anything with the engine running, hydraulics under pressure, work beneath a raised implement, thresher and chaff-cutter blade areas, grain-bin entry, electrical panels on pump sets. For these the app shows the guard-and-stop message and the dealer's number.

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3. The five MVP categories

#	Category	Reach	Guided fixes	Money recovered
1	Mobile phone + charger	97% of households	Storage, charging port, cable, network and SIM, stuck update, battery health	Battery and defect warranty, spare-part price
2	TV + DTH set-top box	61% rural / 79% urban	Input, cable, remote pairing, re-scan, recharge, blink codes	Panel failure in warranty, disputed visit charges
3	Refrigerator	33% rural / 68% urban	Thermostat, coils, door seal, defrost drain, clearance	Compressor cover, usually far longer than the unit warranty
4	Cooling — air cooler and AC	Largest home-service demand	Pads, pump, water, fan; filter, mode, drain, error code, breaker	Compressor warranty, AMC upselling
5	**Two-wheeler**	59% rural / 68% urban own a motor vehicle	Warning lights, won't-start checklist, fuse, bulb, chain, tyre pressure, battery terminals	Free-service schedule, warranty, disputed labour

Evidence base unchanged from version 1: HCES 2023-24 via the EAC-PM paper for ownership; the Department of Consumer Affairs Repairability Index committee for the mobile complaint series (19,057 → 21,020 → 22,864 across 2022-23 to 2024-25); the Right to Repair portal for the government's own priority sectors, which include farming equipment and automobiles; the team's 34-category research model for fault rates and DIY share.

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4. Voice interface — hands-free mode

Voice is not an accessibility extra in this product. On a roadside or in a field, it is the only usable interface.

Behaviour

- One toggle turns on hands-free mode. From then on every step is spoken, and the app listens between steps.
- Commands are short and dialect-tolerant: next, repeat, back, stop, done, not working, help.
- Barge-in: the user can interrupt playback at any moment.
- Every step's safety line is spoken **before** the instruction, never after.

- The emergency interrupt listens on the voice channel too. Spoken words for smell, leak, spark, smoke, burning or shock stop the session and switch to the emergency script.
- Push-to-talk is always available as the fallback when noise defeats continuous listening.
- The screen keeps working in parallel — voice is a mode, not a separate product.

Implementation

Browser speech APIs first, because they cost nothing and work offline on many devices. A cloud speech service is the fallback where Indic recognition quality is weak; evaluate Bhashini, the Indian government's language-technology platform, alongside commercial providers, and decide on measured word-error rate for Hindi in noise, not on marketing claims. Cache the spoken audio for the current case so a dropped connection does not silence the app mid-repair.

Test it the way it will be used. Evals must include recordings made next to a running engine, in wind, and with two people talking.

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5. Revised MoSCoW — six MUSTs

Version 1 had twelve. That is not an MVP; it is a list that invites exactly the critique you anticipated. Six features carry the entire promise, and everything else can arrive later without breaking the story.

MUST — six

#	Feature	What it includes	Why it cannot be cut
M1	**Voice-first intake, with photo**	Hands-free mode, spoken symptom description, photo of the label, error code, dashboard light or bill, with a confirm-and-edit screen	The user cannot type, and often cannot use their hands
M2	**Identify**	Category and model family from the photo or the spoken description, then the specific fault	Everything downstream depends on knowing what the thing is
M3	**Safety gate**	Risk class on every step, per-category DIY ceiling, hard-stop list, emergency interrupt on text, voice and photo	The product's licence to operate. Never optional
M4	**Guided fix**	One step per screen, one picture per step, spoken aloud, "did it work?" branching, escalation after about five steps	This is the product
M5	**Who-pays verdict**	Deterministic rules with a cited source per right: warranty, statutory rights, free-service schedule	The differentiator. Without the citation it is just an opinion
M6	**Outcome pack**	Complaint letter or mechanic brief as a PDF, plus a shareable case link — which also gives helper mode for free	The user must leave with something in hand

Engineering non-negotiables — not features, but the definition of done

These were MUSTs in version 1 and did not belong there. They are how we build, and they take no product scope:

- Golden test cases per category, plus **red-team refusal cases**. A release fails if the app produces a step for a gas leak, a regulator, an opened inverter, a brake line or a lithium top-up.
- Every rights answer carries a source link and a last-checked date, enforced in code.
- A minimal event log: fixed, claimed, brief, abandoned.

SHOULD — after the MVP earns it

WhatsApp as an entry channel · full Hindi UI rather than Hindi answers only · spare-part price and manual deep links through the Right to Repair portal · warranty-expiry and follow-up reminders · offline pack for the five categories · outcome analytics beyond the minimal log.

COULD

Phase 2 and 3 categories as listed in section 2 · more Indian languages · insurance policy reader · recall and standards check · verified local technician hand-off · bills pulled from email or cloud storage.

WON'T — this release, stated on the slide

Gas repair steps · mains wiring · high-voltage, EV or sealed units · refrigerant · brakes, airbags and fuel systems · working under a raised vehicle or implement · industrial and workplace equipment · filing a consumer case on the user's behalf · payments · medical and assistive devices · markets outside India.

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6. What is *not* RAG

A point the team should be clear about before building the corpus: **most of the product's accuracy comes from structured data, not from retrieval.** Three things stay in tables and code, never in the vector store:

1. **Error and fault code tables** — brand, model family, code, meaning, first-line steps, DIY ceiling, source. Exact lookup, no embedding.
2. **Rules** — warranty terms, statutory rights, free-service schedules, as versioned JSON with a source URL and last-checked date on every rule.
3. **Safety** — risk classes, hard stops, emergency scripts. Hand-written, reviewed, never generated.

RAG covers the long tail: the manual paragraph that explains what a specific model does, the warranty clause in a specific brand's booklet, the safety instruction in an operator's manual.

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7. The RAG corpus — exactly what to collect

Per document, the ingestion pipeline records: `doc_id · category · brand · model_family · doc_type · jurisdiction · language · edition or year · source_url · last_checked · licence · section or page · risk_class`. Retrieval filters on model family and category **before** semantic search, and no answer is returned without a citation.

7.1 MVP corpus — phase 1, five categories

A. Product documentation (the bulk of the work)

Document	Detail	Where from
User manuals	Top 5 brands × top 2–3 model families per category	Brand support sites; Right to Repair portal
Service and installation manuals	Where publicly available; they carry the fault trees	Brand support sites
Error-code appendices	Extracted separately and promoted into the structured tables	From the manuals above
Remote-control and quick-start guides	Critical for AC and TV, where the fault is usually a setting	Brand sites
Warranty booklets and terms	Brand-level, including compressor and battery exceptions	Brand sites; product box scans
Extended-warranty and AMC terms	The documents behind the upselling the app challenges	Retailer and brand sites
Spare-part price and availability lists	Also feeds the "is this worth repairing" answer	Right to Repair portal

B. Rights and redress

Consumer Protection Act 2019 (relevant chapters on defect, deficiency and unfair trade practice) · Consumer Protection (E-Commerce) Rules 2020 · Legal Metrology (Packaged Commodities) Rules, for declared warranty and MRP · the National Consumer Helpline process and 1915 · the e-Jagriti / consumer commission filing procedure, jurisdiction thresholds, fees and limitation period · a small set of consumer-commission orders on defective goods and deficiency in service, used as precedent examples only, never as the source of a right · the Right to Repair portal's own commitments per sector.

C. Two-wheeler documents (new, and the most valuable per page)

Owner's manual per model family · service booklet with the free-service schedule and intervals · warranty booklet including the battery and consumables exclusions · **dashboard telltale and warning-light glossary**, which is what the photo lookup answers against · fuse-box layout and rating table · bulb-type table · tyre pressure placard values and the tyre-marking guide · recommended lubricant and chain-slack specifications · the manufacturer's recall notices · the standard motor insurance policy wording, own-damage claim process and what a zero-depreciation cover changes · for phase 2 cars, the generic OBD-II diagnostic trouble code dictionary plus manufacturer-specific code lists.

D. Safety corpus

The safety and warning sections extracted from every manual above, tagged `risk_class` · brand "do not" lists · the LPG safety guidance, the oil marketing companies' public liability and personal accident policy, the five-yearly inspection rule and the 1906 emergency number · inverter and battery handling guidance from the battery manufacturers · electrical safety basics for the household categories.

E. Field knowledge

Verified brand service-centre directories and authorised-workshop lists, so the app never sends a user to a number found in a search result · public service bulletins and known-issue notices · repair-community datasets such as the Open Repair Alliance records, used for frequency and success rates rather than for instructions.

Indicative volume for the MVP: roughly 60–90 documents, of which about 50 are manuals; in the order of 5,000–15,000 chunks. This is a two-to-three-week content effort for the non-engineers, and it is

the critical path.

7.2 Phase 3 — agriculture corpus, listed now so nobody starts from zero

Tractor operator's manuals and service schedules for the dominant model families · implement manuals: rotavator, cultivator, seed drill, thresher, chaff cutter · knapsack and power-sprayer manuals with nozzle and calibration charts · drip and micro-irrigation design and maintenance guides · pump-set and starter manuals · farm-machinery safety guidance from the agricultural research and training institutes · machinery test reports from the farm machinery training and testing institutes · state agriculture department advisories in the local language · subsidy, warranty and dealer-obligation documents.

7.3 Licensing — raise this before ingestion, not after

OEM manuals are copyrighted. The defensible pattern is: retrieve, quote a short passage with attribution, and always link to the manufacturer's own page or the Right to Repair portal rather than redistributing the file. Community repair content usually carries a share-alike, non-commercial licence, which conflicts with a commercial product — check each source's terms and record the licence in the document metadata. Put this in the deck; a jury that asks about it and gets a clear answer will trust everything else more.

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8. Technology decision

Claude as the coding assistant and the runtime model — endorsed. The team has one engineer and six non-engineers, so the assistant's job is leverage, not substitution: it multiplies the engineer's output and lets the other six contribute content, rules and test cases without writing production code. At runtime the same model family covers three jobs — reading a photographed label or dashboard, drafting the letter, and speaking the steps. It decides nothing; the rules engine does that.

Build recommendation, revised for a team with an engineer. With an engineer owning the codebase, the argument for a managed sandbox weakens and the argument for a normal repository gets stronger. The recommendation is:

- **Primary: an owned Next.js repository on GitHub, deployed to Vercel, with Supabase for data.** The engineer owns the repo, reviews what the assistant writes, and controls the deployment. This avoids the two constraints that would otherwise shape the project — a publishing region that cannot be changed after the first publish, and effort-based agent billing — and it is the stack the product would keep after the capstone.
- **Replit stays, in a defined role: the sandbox for the six non-engineers.** It is genuinely good for this — no local setup, a live URL, a place to try a screen or a prompt and show it to the team the same hour. The rule is that nothing durable lives there: experiments graduate into the repo through a pull request the engineer reviews.
- **If the team still prefers to build and demo entirely on Replit**, that remains defensible, with three conditions: choose the publishing geography deliberately before the first publish, because it cannot be changed afterwards and there is no India region; give one person the account and a weekly spend ceiling, because agent billing is effort-based; and keep durable data in Supabase rather than the built-in database, so a later move is a redeploy rather than a rewrite.

Alternatives. Lovable and Bolt are fastest to a pretty screen and weakest where our value sits — rules engine, retrieval, evals. Firebase Studio adds a second vendor's auth and data model for no gain. The Cursor-or-Claude-Code plus Vercel plus Supabase path, which version 1 deferred as "needs an engineer", is now the primary recommendation precisely because the team has one.

Layer	Choice
Build	Next.js repo on GitHub, owned and reviewed by the engineer; Claude Code as the assistant; Replit as the non-engineers' sandbox
App	Next.js TypeScript as an installable PWA, camera and microphone in the browser
Hosting	Vercel, with the Replit deployment kept as a demo-day fallback
Voice	Browser speech APIs first; a cloud Indic speech service as fallback, chosen on measured word-error rate in noise
Runtime model	Claude behind a thin provider-agnostic wrapper with JSON-schema outputs; a fast model for extraction, a stronger one for diagnosis and drafting
Orchestration	Explicit state machine: intake → identify → **safety gate** → guided fix → verdict → outcome pack
Rules and safety	Versioned JSON plus plain functions; every rule carries its source and last-checked date
Data and retrieval	Supabase: Postgres, pgvector, storage, anonymous auth, row-level security
Content pipeline	Offline ingestion scripts with the metadata schema in section 7
Documents	Server-side PDF for letters and mechanic briefs
Evaluation	Golden and red-team cases in version control, run before every release

Compliance. Bills, serial numbers, registration details and photographs are personal data under the DPDP Act 2023: consent, case deletion, no training on user uploads, stated data region. Every legal output carries its source and an "information, not legal advice" line; every safety output carries the authorised service number.

9. Delivery and risk

Workstream	Owner	Output
Architecture, repo, deployment, code review	The engineer	The production path; nothing merges without review
App, voice, retrieval, rules engine, PDF	The engineer with one or two assistant-supported builders	The product
Content: manuals, codes, warranty terms for five categories	Non-engineers, one category owner each	The critical path
Rights rules, India	The team member reading consumer law	Versioned JSON with sources
Safety: risk classes, hard stops, emergency scripts, red-team cases	One owner, whole-team review	The claim we make on stage

Workstream	Owner	Output
Evals and metrics	Scrum master	Weekly score as the definition of done

Definition of done for a category: ten faults documented with sources, golden cases passing, red-team cases refused, one cited verdict, one letter and one mechanic brief generated end to end, and the whole flow completed once in hands-free mode.

Risk	Mitigation
Content runs out of time	Five categories, ten faults each, hard cap. No sixth category before demo day
One engineer becomes the bottleneck	Keep the engineer on architecture, the safety gate and review; route UI and content work to assistant-supported builders; nothing that blocks the demo sits in one person's queue unreviewed
Voice fails in a noisy venue	Push-to-talk fallback, rehearsed; recorded backup of the hands-free act
A wrong safety answer on stage	Red-team cases block the release; the refusal is demonstrated deliberately
Vehicle advice liability	Narrow allowed list, explicit refusals, roadside assistance for anything else
Manual licensing challenged	Link rather than redistribute; licence recorded per document
Sandbox work diverging from the repo	Experiments graduate through a reviewed pull request; nothing durable lives in the sandbox

10. Demo day

1. **Quick win — TV.** Wrong input, "No Signal", fixed in under a minute.
2. **Hands-free — two-wheeler.** The scooter won't start. The rider's hands are on the machine, so the whole fix runs by voice: kill switch, side stand, fuel, then the fuse. Ends with what the garage would have charged.
3. **Money — refrigerator.** Four years old, not cooling. Plate, warranty card and bill photographed; safe checks; a cited "who pays" verdict; a complaint letter drafted in Hindi.
4. **The refusal — LPG.** "Gas smell from the cylinder." The app abandons diagnosis, runs the evacuation script, gives 1906. No gas on stage, ever.

Sources

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